Worksheet 14 – Hopkins statistic to assess cluster tendency

Instead of a multi-dimension dataset, let's use a simple 1-dimension dataset and compute Hopkins statistic. Suppose your data is made up of 10 numbers: 1, 1.2, 1.4, 21, 20.2, 19.8, 21, 21.2, 20, 21

(a) Sketch on a number line the 10 points to convince yourself there are obviously 2 clusters that are well separated with no overlap

(b) Based on your answer in (a), would you expect Hopkins statistic to be low (close to 0), about 0.5 or high (above 0.75 to 1)?

(c) Based on your understanding of the algorithm behind calculations of the Hopkins statistic, would you expecct Hopkins statistic H to change each time you run the R command hopkins()? Assume you did not use set.seed().

(d) Use R's clustertend package and the hopkins() function to obtain the value of Hopkins statistic. Run hopkins() a few times to validate your answer in part (c). Write your code here:

(e) Now instead of using clustertend package, use R's hopkins package and the hopkins() function to obtain the value of Hopkins statistic. How do the values of H compare (any difference between clustertend and hopkins packages)? Write code and answer below:

(f) If you had used a regularly spaced dataset, what value of Hopkins statistic would you expect? It is best that students make use of H = sum(y)/ [sum(x) + sum(y)] to convince themselves of the answer.

(g) Validate part (f) using R's hopkins{} package to confirm your understanding. Note that clustertend package is deprecated (this means it is available but its use is not recommended – your answer to part (e) helps to understand why hopkins package is preferred). Write code here:

(h) Instead of regularly spaced data, use data that are drawn from a random uniform distribution with points that are scattered over the number line (say from smallest value of 1 to maximum value of 20). In this case, what value of value of Hopkins statistic would you expect? Students should realize runif() would be appropriate to generate such data. It is best that students make use of H = sum(y)/ [sum(x) + sum(y)] to convince themselves of the answer.

(i) Validate part (h) using R's hopkins package. Write code here: